

CYBERSECURITY REPORT

Name: Akash B Nair

B. Tech CSAI

1. Introduction

During my cybersecurity internship, I worked in a controlled lab environment to learn basic network security concepts and vulnerability assessment techniques. The objective was to gain hands-on experience with tools such as Nmap, Wireshark, and Nessus while following ethical practices.

2. Environment Setup

1. Tools Used:
 - Nmap
 - Wireshark
 - Nessus
 - Metasploitable2
 - Kali Linux
2. OS/VM:
 - Software: VirtualBox
 - Target machine: Metasploitable2
 - Attacker machine: Kali Linux

3. Reconnaissance

1. Target IP: 192.168.56.101
2. Open ports and running services:
 - 21/FTP
 - 22/SSH
 - 23/telnet
 - 25/SMTP
 - 53/DNS
 - 80/HTTP
 - 111/RPCBind
 - 139/NetBIOS-SSN
 - 445/SMB

- 512/exec
- 513/login
- 514/shell
- 1099/Java RMI
- 1524/Bind Shell
- 2049/NFS
- 2121/FTP
- 3306/MySQL
- 5432/PostgreSQL
- 5900/VNC
- 6000/X11
- 6667/IRC
- 8009/AJP13
- 8180/HTTP

Total open ports: 23 TCP ports

3. Potential vulnerabilities:

- FTP is outdated
- Telnet is enabled
- Samba services are older versions
- MySQL and PostgreSQL are exposed on network

4. Network Traffic Analysis

- Plain-text FTP credentials were observed in Wireshark.
- HTTP and FTP traffic were successfully analysed.
- TCP SYN packets were identified during connection establishment.
- No suspicious activity was detected in the lab environment.

5. Threat Summary & Impact

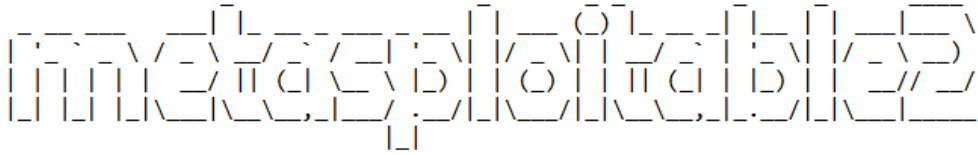
Threat	Description	Potential Impact	Mitigation
FTP Service	FTP transmits usernames and passwords in plain text	Credential theft	Replace FTP with SFTP/FTPS
Weak Web Security	DVWA does not use HTTPS encryption	Unauthorized access and data exposure	Enable HTTPS and strong authentication
Open Telnet Port	Telnet communication is unencrypted	Sensitive information leakage	Disable Telnet and use SSH
Multiple Open Services	Several services were running on the target machine	Increased attack surface	Disable unnecessary services and keep software updated

6. Conclusion

This internship provided practical exposure to cybersecurity tools and improved my understanding of vulnerability assessment and network analysis.

7. Evidence and Screenshots

Metasploitable2 Home Page Accessed Successfully



Warning: Never expose this VM to an untrusted network!

Contact: [msfdev\[at\]metasploit.com](mailto:msfdev[at]metasploit.com)

Login with msfadmin/msfadmin to get started

- [TWiki](#)
- [phpMyAdmin](#)
- [Mutillidae](#)
- [DVWA](#)
- [WebDAV](#)

Figure: Metasploitable2 Home Page Accessed Successfully

Wireshark HTTP Traffic Capture (GET / and 200 OK)

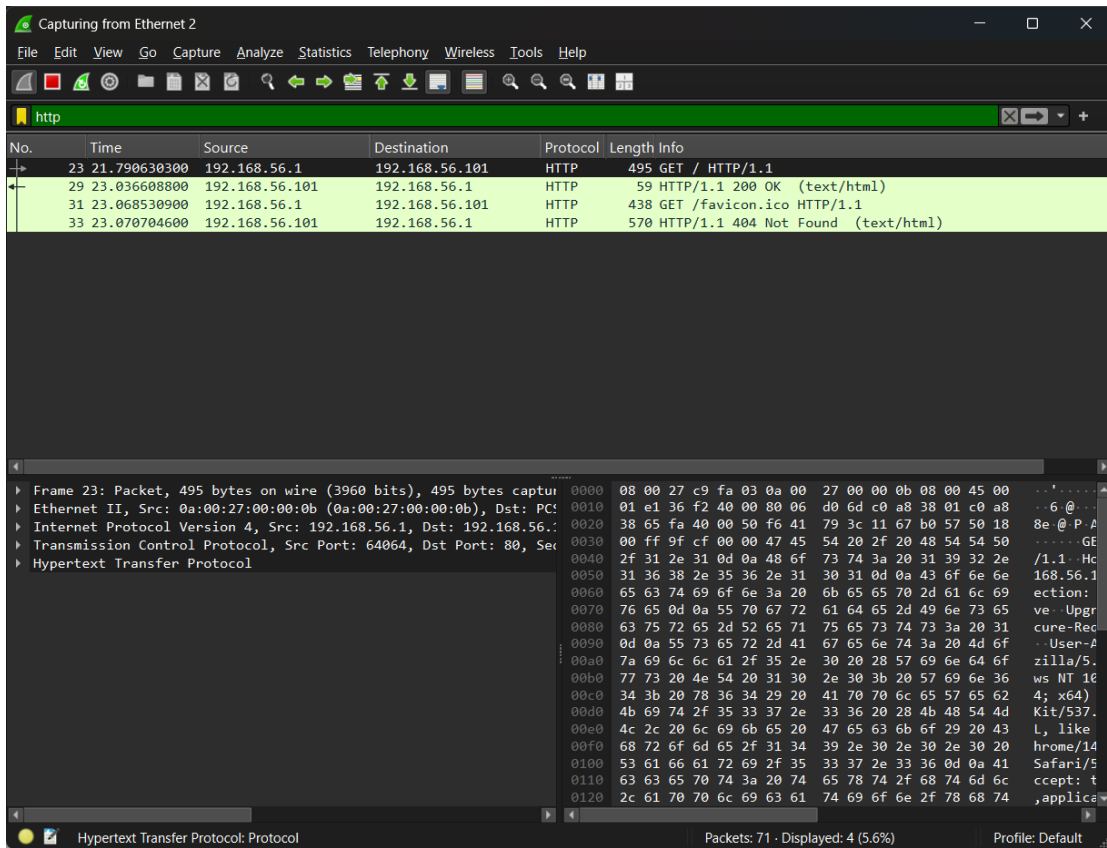


Figure: Wireshark HTTP Traffic Capture (GET / and 200 OK)

Follow HTTP Stream Analysis



Figure: Follow HTTP Stream Analysis

FTP Plaintext Credentials in Wireshark

The image shows a Wireshark capture of FTP traffic. The top pane displays a list of packets, with packet 465 selected. The middle pane shows the details of the selected packet, highlighting the 'File Transfer Protocol (FTP)' section. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
456	526.182394200	192.168.56.101	192.168.56.1	FTP	74	Response: 220 (vsFTPd 2.3.4)
457	526.184342800	192.168.56.1	192.168.56.101	FTP	68	Request: OPTS UTF8 ON
459	526.212637200	192.168.56.101	192.168.56.1	FTP	80	Response: 200 Always in UTF8 mode.
461	536.293113300	192.168.56.1	192.168.56.101	FTP	69	Request: USER msfadmin
462	536.302367900	192.168.56.101	192.168.56.1	FTP	88	Response: 331 Please specify the password.
464	542.973587300	192.168.56.1	192.168.56.101	FTP	69	Request: PASS msfadmin
465	543.014669200	192.168.56.101	192.168.56.1	FTP	77	Response: 230 Login successful.

Frame 456: Packet, 74 bytes on wire (592 bits), 74 bytes captured
Ethernet II, Src: PCSSystemtec_c9:fa:03 (08:00:27:c9:fa:03), Dst: 00:0c:35:dd:40:00
Internet Protocol Version 4, Src: 192.168.56.101, Dst: 192.168.56.1
Transmission Control Protocol, Src Port: 21, Dst Port: 8723, Seq: 38010015
File Transfer Protocol (FTP)
[Current working directory:]

Figure: FTP Plaintext Credentials in Wireshark

TCP SYN Packet Filter Results

Wireshark packet capture showing a TCP SYN flood attack. The packet list displays multiple SYN packets from 192.168.56.1 to 192.168.56.101. The packet details for frame 451 show an Ethernet II frame, an Internet Protocol Version 4 packet, and a Transmission Control Protocol SYN packet.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.56.1	192.168.56.101	TCP	66	1043 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK
2	0.000401000	192.168.56.1	192.168.56.101	TCP	66	25713 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK
42	123.282981300	192.168.56.1	192.168.56.101	TCP	66	55266 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK
43	123.283334800	192.168.56.1	192.168.56.101	TCP	66	11297 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK
62	124.017113800	192.168.56.1	192.168.56.101	TCP	66	41566 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK
111	151.797741000	192.168.56.1	192.168.56.101	TCP	66	20232 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK
126	152.116949100	192.168.56.1	192.168.56.101	TCP	66	58604 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK
164	192.547293500	192.168.56.1	192.168.56.101	TCP	66	54997 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK
192	196.255354700	192.168.56.1	192.168.56.101	TCP	66	42278 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK
229	224.350542900	192.168.56.1	192.168.56.101	TCP	66	14207 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK
313	236.514694000	192.168.56.1	192.168.56.101	TCP	66	49669 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK
321	236.523029600	192.168.56.1	192.168.56.101	TCP	66	50263 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK
451	526.153117200	192.168.56.1	192.168.56.101	TCP	66	8723 → 21 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=1 SACK

Frame 451: Packet, 66 bytes on wire (528 bits), 66 bytes captured
 Ethernet II, Src: 0a:00:27:00:00:0b (0a:00:27:00:00:0b), Dst: PC:
 Internet Protocol Version 4, Src: 192.168.56.1, Dst: 192.168.56.1
 Transmission Control Protocol, Src Port: 8723, Dst Port: 21, Seq:

Packets: 467 · Displayed: 13 (2.8%)

Figure: TCP SYN Packet Filter Results